

A Computationally-Integrated Simulation-Focused Problem-Based Learning Framework for Engineering Dynamics: Design, Implementation, and Mixed-Methods Evaluation in an OBE-Aligned Undergraduate Course

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2 ABSTRACT

3 Engineering dynamics courses demand simultaneous development of mathematical modelling,
4 computational implementation, and communication competencies, yet conventional instruction
5 rarely provides authentic contexts for their integration. The primary objective of this study was to
6 design, implement, and empirically evaluate a Simulation-Focused Problem-Based Learning (SF-
7 PBL) framework that integrates first-principles Newtonian and Lagrangian mechanics with Python-
8 based numerical simulation within an Outcome-Based Education (OBE)-aligned curriculum.
9 The framework was embedded within a 15-week undergraduate Dynamic System Modeling
10 (DSM) course ($N = 53$, B.Tech Mechatronics Engineering, Semester IV, January–April 2026)
11 at a single NBA-accredited institution in India. It operationalises a four-phase computational
12 pipeline—System Identification, Mathematical Modelling, Computational Simulation, Validation
13 & Communication—scaffolded through an Explain-Try-Challenge (E-T-C) model and assessed
14 via a six-judge external expert panel. A pre–post mixed-methods design measured self-efficacy
15 (10-item Likert scale), conceptual knowledge (5-item MCQ), PBL satisfaction, AI tool usage, and
16 qualitative breakthrough reflections. Self-efficacy improved significantly (pre: $M = 2.92$, $SD = 0.21$;
17 post: $M = 3.80$, $SD = 0.40$; $t(52) = 13.91$, $p < .001$, Cohen's $d = 2.72$, pooled SD). Knowledge gains
18 were large ($t(52) = 11.32$, $p < .001$, $d = 1.20$; Hake's $g = 0.50$, medium interactive-engagement
19 band). All ten self-efficacy items improved significantly ($p < .001$). PBL satisfaction was high
20 ($M = 3.74$, $SD = 0.30$). Eighty-five percent of students used AI tools, primarily for computational
21 coding, while mathematical derivation remained predominantly human-led. Qualitative themes
22 identified simulation validation, PID tuning, and linking derivation to computation as commonly
23 reported breakthrough moments. Exhibition rubric scores from 17 student projects evaluated by
24 the six-judge panel ($M = 82.62/100$, $SD = 9.98$) confirmed strong authentic performance. The
25 SF-PBL framework offers a replicable, empirically-evaluated model for computationally-integrated
26 dynamics pedagogy in OBE-aligned STEM programmes.

27 **Keywords:** problem-based learning; engineering dynamics; computational simulation; self-efficacy; OBE; authentic assessment;
28 AI-assisted learning; STEM education

1 INTRODUCTION

29 1.1 The Pedagogical Challenge in Engineering Dynamics

30 Engineering dynamics—encompassing kinematics, constitutive laws, Lagrangian mechanics, and vibra-
31 tion analysis—occupies a foundational position in mechanical and mechatronics engineering curricula
32 worldwide. Despite its centrality, the course presents persistent and well-documented pedagogical chal-
33 lenges: the mathematical formalism is demanding, physical intuition is difficult to develop, and the
34 connection between abstract theory and real engineering systems is rarely made explicit within traditional
35 lecture-driven instruction (Streveler et al., 2008). Withdrawal, failure, and incomplete (DFW) rates in core
36 dynamics and mechanics courses at undergraduate level remain a persistent concern, with some institutions
37 reporting DFW rates exceeding 20–30% in introductory engineering mechanics sequences (Crisp et al.,
38 2009). Students frequently navigate these courses through surface-level pattern-matching without building
39 the conceptual depth required for engineering practice.

40 Compounding this challenge is the increasing centrality of computational tools in modern engineering
41 practice. Graduates are expected not only to derive equations of motion analytically but to implement them
42 numerically, simulate dynamic system behaviour across time domains, and interpret computational outputs
43 as engineering evidence—a set of competencies that conventional closed-book assessments rarely evaluate
44 authentically (Justo et al., 2021). The emergence of AI-assisted coding tools (ChatGPT, GitHub Copilot,
45 Claude Sonnet) has further transformed the competency landscape: students can now generate working
46 simulation code in seconds, fundamentally shifting the pedagogical question from “*can you write the code?*”
47 to “*do you understand what the code is computing and why?*” (Tlili et al., 2023).

48 1.2 Problem-Based Learning in Engineering: Promise and Gap

49 Problem-Based Learning (PBL), as defined by Hmelo-Silver (2004), places authentic, ill-structured
50 problems at the centre of learning, requiring students to self-direct their inquiry, build domain knowledge
51 collaboratively, and communicate solutions to real audiences. In engineering education, PBL has consistently
52 demonstrated improvements in conceptual understanding, motivation, and collaborative problem-solving
53 competencies compared to conventional instruction (Prince and Felder, 2006; Dym et al., 2005; Dochy
54 et al., 2003). Freeman et al. (2014) meta-analysed 225 studies and found that active learning produced a
55 mean Cohen’s $d = 0.47$ improvement in examination performance relative to traditional lecturing, with
56 engineering courses showing the strongest effect.

57 However, three specific implementation gaps persist in the published literature on PBL in engineering
58 dynamics and related computationally intensive courses. First, most PBL implementations position a
59 physical prototype or design report as the primary learning artefact—rarely is mathematical derivation from
60 first principles and its computational simulation verification foregrounded as the core intellectual product.
61 Second, authentic assessment in PBL contexts typically relies on faculty panels or peer evaluation, with
62 multi-domain external expert panels covering industry, intellectual property, and communication remaining
63 uncommon. Third, while many institutions operate under Outcome-Based Education (OBE) frameworks
64 mandated by accreditation bodies such as NBA (India) and ABET (USA), explicit and auditable traceability
65 from Course Outcomes through Bloom’s taxonomy levels to rubric dimensions is rarely demonstrated
66 empirically.

These gaps distinguish the present work from three established pedagogical models. Classical PBL, as originally operationalised in medical education (Hmelo-Silver, 2004), centres on open-ended clinical reasoning with no structured computational pipeline; the primary learning artefact is a diagnostic report, not a first-principles mathematical derivation verified through simulation. Project-based and capstone-design approaches (Dym et al., 2005) typically foreground physical prototyping, treating mathematical modelling as instrumental rather than as the core intellectual product. Standalone computational modelling courses (Justo et al., 2021) typically pre-specify the governing equations, asking students to implement rather than derive them. SF-PBL integrates these elements into a single cohesive framework where student-led derivation, computational simulation, and authentic public defence form an interdependent sequence. Recent work on cross-disciplinary team formation in computationally-intensive XR engineering courses at the same institution (Nanade and Anne, 2026) further underscores the pedagogical value of structured active-learning interventions in engineering curricula.

1.3 Research Questions

This study addresses these gaps through the design, implementation, and empirical evaluation of a Simulation-Focused Problem-Based Learning (SF-PBL) framework. Three research questions guided the study:

- RQ1:** Does the SF-PBL framework produce statistically significant and practically meaningful gains in students' self-efficacy for dynamic system modelling competencies?
- RQ2:** Does the SF-PBL framework produce statistically significant gains in conceptual knowledge of dynamics principles?
- RQ3:** How do students integrate AI tools within the SF-PBL structure, and what qualitative experiences characterise their learning breakthroughs?

1.4 Contributions

This paper makes four original contributions to the engineering education literature: (1) a documented, replicable four-phase SF-PBL computational algorithm with full CO→Bloom's→NBA Programme Outcome traceability; (2) an authentic assessment model using a six-domain external expert panel; (3) empirical evidence of large-to-very-large pre-to-post learning gains ($d > 1.0$) from a mixed-methods study within a single cohort ($N = 53$); and (4) a characterisation of emergent human–AI complementarity within a structured PBL framework.

2 THEORETICAL FRAMEWORK

2.1 Simulation-Focused Problem-Based Learning (SF-PBL)

SF-PBL is defined in this study as a variant of PBL in which **mathematical derivation from first principles and its subsequent computational simulation constitute the primary learning artefact**. This distinguishes SF-PBL from conventional project-based learning (where the product is a physical prototype or design report) and from standalone computational modelling courses (where system identification and mathematical modelling steps are typically pre-specified by the instructor rather than student-led).

The SF-PBL framework rests on three interlocking theoretical pillars. *First*, **Hmelo-Silver's PBL cycle** (2004): students encounter an authentic, ill-structured engineering system (student-selected), engage in

self-directed inquiry to identify governing physics, construct knowledge through mathematical and computational means, and reflect collaboratively on computational outcomes and their physical interpretations. *Second, Bandura's self-efficacy theory* (1997): mastery experiences—successfully deriving equations of motion and observing their agreement with simulation output—constitute the most potent mechanism for building domain-specific self-efficacy. The four sequential SF-PBL phases provide escalating mastery experiences mapped directly to Course Outcomes, ensuring that self-efficacy is built incrementally and is domain-specific rather than generalised. *Third, Hake's interactive-engagement framework* (1998): courses that require active student engagement with phenomena—rather than passive reception of lecturer-transmitted information—systematically outperform traditional instruction on normalised conceptual gain. SF-PBL's four-phase structure, requiring active derivation, numerical implementation, and public defence of results, is designed to achieve medium-to-high interactive-engagement gain.

Underpinning these three pillars is a constructivist epistemology rooted in Vygotsky's concept of the Zone of Proximal Development (ZPD) (Vygotsky, 1978): the E-T-C instructional model (§2.3) provides dynamic scaffolding within students' ZPD, systematically reducing support as competence develops across the 15-week course. Kolb's (1984) experiential learning cycle—Concrete Experience → Reflective Observation → Abstract Conceptualisation → Active Experimentation—maps directly onto the four SF-PBL phases, providing a further theoretical grounding for the sequential pipeline design.

Table 1 summarises the alignment between these theoretical foundations, the SF-PBL phases, and the assessment instruments employed.

Table 1. Theoretical–Pedagogical Synthesis: Mapping of theoretical foundations to SF-PBL phases, E-T-C stages, and assessment instruments.

Theoretical Pillar	SF-PBL Phase	E-T-C	Assessment
Hmelo-Silver PBL Cycle	Phase 1: System ID	Explain	Project proposal
Bandura Self-Efficacy (Mastery)	Phases 2–3: Derivation + Sim.	Try	SE scale (pre–post)
Hake Interactive Engagement	Phases 3–4: Sim. + Validation	Challenge	MCQ + Exhibition
Vygotsky ZPD / Scaffolding	All phases (fading support)	E-T-C	Reflection journals
Kolb Experiential Learning	Phase 4: Exhibition	Challenge	6-judge panel

2.2 OBE Framework and NBA Alignment

The DSM course was designed using OBE principles (Spady, 1994) with explicit backward design from NBA Programme Outcomes (POs) to Course Outcomes (COs) to assessment rubric dimensions. Five COs were defined (Table 2), each assigned a Bloom's taxonomy level and mapped to the relevant NBA PO.

Table 2. Course Outcomes, Bloom's Taxonomy Levels, and NBA Programme Outcome Mapping

CO	Statement	Bloom's	NBA PO
CO1	Apply kinematic analysis to multi-body systems	Apply (L3)	PO1, PO2
CO2	Construct FBDs for complex dynamic mechanisms	Analyse (L4)	PO1, PO2
CO3	Derive equations of motion using Newton's laws	Analyse (L4)	PO1, PO2
CO4	Apply Lagrangian mechanics to holonomic systems	Evaluate (L5)	PO1, PO2
CO5	Implement ODE solvers in Python and validate simulations	Create (L6)	PO2, PO5

PO1 = Engineering Knowledge; PO2 = Problem Analysis; PO5 = Modern Tool Usage.

In the NBA/OBE framework, Programme Outcome (PO) attainment is evidenced through Course Outcome performance. COs 1–4 map jointly to PO1 (Engineering Knowledge) and PO2 (Problem Analysis), assessed through the mathematical derivation and simulation phases of SF-PBL. CO5 additionally maps to PO5 (Modern Tool Usage), assessed through the Python implementation requirement. The six-judge exhibition rubric spans all five COs, with each judge dimension providing evidence for one or more POs: the Mathematics judge (NG) assesses CO1–CO4 directly, while the Industry 4.0 judge (MK) assesses CO5 through evaluation of computational implementation quality. Course Outcome attainment is analysed in Section 4.6 using both direct (MCQ) and indirect (self-efficacy) assessment evidence.

2.3 The Explain-Try-Challenge (E-T-C) Instructional Model

Each 90-minute lecture session followed a three-phase active-learning structure (Prince and Felder, 2006): **Explain** (30 min)—conceptual introduction with worked examples on the whiteboard; **Try** (40 min)—guided structured practice using Jupyter notebooks with worked-example scaffolds; **Challenge** (20 min)—an open-ended, unseen problem connecting the lecture content to the student's own SF-PBL project system, requiring them to apply the newly introduced concept to their self-selected engineering system. The Challenge phase operationalised Vygotsky's ZPD: it consistently targeted skills just beyond students' independent capability, making the weekly lecture a direct scaffold for the longer-term SF-PBL project.

3 MATERIALS AND METHODS

3.1 Research Design

This study employed a **single-cohort pre-post mixed-methods design** (Creswell and Creswell, 2018), integrating: (a) a quantitative strand comprising repeated-measures survey data (self-efficacy, knowledge), exhibition rubric scores, and categorical AI usage data; (b) a qualitative strand comprising thematically coded open-ended reflections; and (c) triangulation, in which quantitative gains are corroborated by qualitative breakthrough narratives referencing specific SF-PBL phases. The absence of a concurrent control group is acknowledged as a design limitation (see Section 5.7); however, the single-cohort design was necessitated by the course being a mandatory core subject for the entire Semester IV cohort, precluding random assignment.

3.2 Course Context and Participants

The study was conducted in the Dynamic System Modeling (DSM) course (Course Code: 702MH0C023) at NMIMS MPSTME, Mumbai, during Academic Year 2025–26. Course parameters and participant details are presented in Table 3.

Participation in the research survey was voluntary and anonymised; course grades were not affected by survey participation. Informed consent was obtained from all 53 participants prior to pre-survey administration. The two registered students who declined consent are excluded from all analyses; $N = 53$ applies throughout this paper. As this is a single-cohort, single-institution study, findings should be interpreted within this specific institutional and cultural context.

3.3 The SF-PBL Four-Phase Computational Pipeline

The SF-PBL project ran concurrently with the 15-week course and culminated in a public Mini Project Exhibition on 25 April 2026 at NMIMS MPSTME, organised in collaboration with the NMIMS IUCEE Student Chapter (video documentation available via LinkedIn: <https://www.linkedin.com/posts/>

Table 3. Course Context and Participant Summary

Parameter	Value
Course code	702MH0C023 — Dynamic System Modeling
Institution	NMIMS MPSTME, Mumbai (NBA-accredited)
Programme	B.Tech Mechatronics Engineering, Semester IV
Academic year	2025–26
Course duration	2 January – 25 April 2026 (15 working weeks)
Contact hours	3 credits: 2 lectures + 2 practical hours/week
N registered	55 students
N consenting & participating	53 (2 students did not provide informed consent)
N present at exhibition	53 (all consenting participants; 1 registered student absent from exhibition, scored 0, excluded)
Student teams	17 (2–4 members per team)
Engineering project domains	5
Exhibition date	25 April 2026

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The four-phase pipeline is illustrated in Figure 1. Each phase has specific deliverables, COs, and Bloom's levels, as described below. Representative student project artefacts, including simulation code, hand-drawn Free Body Diagrams, and validation reports across the 17 teams, are archived in the repository (<https://github.com/sunny-nanade/SF-PBL-Engineering-Dynamics-2026>).

Phase 1: System Identification (Weeks 1–4). Students independently selected a real engineering system (no prescribed list), identified its degrees of freedom (DOF) and kinematic constraints, and defined coordinate frames and generalised coordinates. *Deliverable:* System description and kinematic sketch. *CO:* CO1; *Bloom's:* Apply (L3).

Phase 2: Mathematical Modelling (Weeks 5–9). Students drew Free Body Diagrams (FBDs) by hand, applied Newton's Second Law or the Lagrangian formulation to derive coupled Equations of Motion (EOMs), and converted the resulting second-order system to first-order state-space form: $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u})$. *Deliverable:* Analytical EOMs and state-space representation. *CO:* CO2, CO3, CO4; *Bloom's:* Analyse–Evaluate (L4–L5).

Phase 3: Computational Simulation (Weeks 10–13). Students implemented the state-space model using Python 3.10 (`scipy.integrate.solve_ivp`), generated time-domain response plots (position, velocity, energy vs. time), and tuned control parameters (e.g., PID gains for controlled systems). Code was version-controlled on GitHub. *Deliverable:* Working simulation with validated plots. *CO:* CO5; *Bloom's:* Create (L6).

Phase 4: Validation & Communication (Weeks 14–15). Students validated their simulations against analytical benchmarks (energy conservation, limit cases, or published solutions), prepared a 20-slide presentation using the institutional template, and defended their work to the six-judge expert panel at the public exhibition. *Deliverable:* Exhibition presentation and validation evidence report. *CO:* CO5; *Bloom's:* Evaluate–Create (L5–L6).

3.4 Student Projects: 17 Teams, 5 Engineering Domains

The 17 student teams were distributed across five engineering project categories (Table 4), classified by degrees of freedom, coupling degree, and control system requirements as Very High (VH, 6 teams), High (H, 6 teams), or Medium (M, 5 teams) complexity.

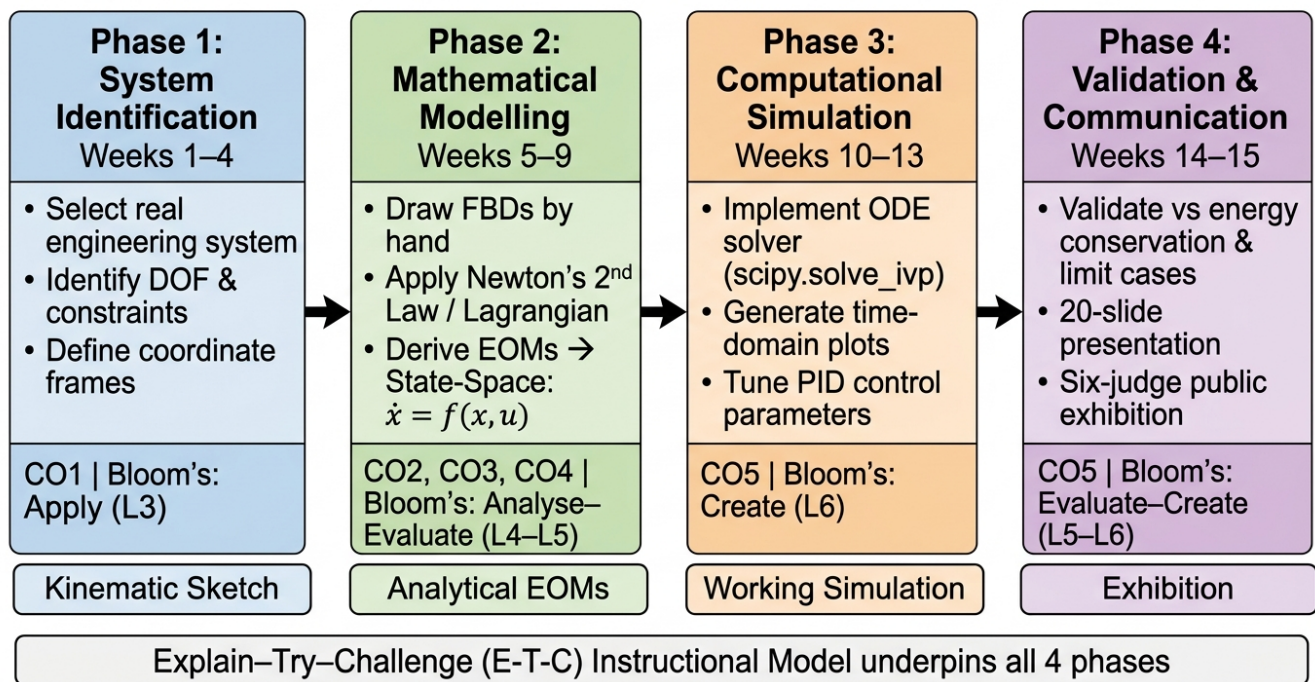
Figure 1. The SF-PBL Four-Phase Computational Pipeline (15-Week Course, N=53)

Figure 1. The SF-PBL Four-Phase Computational Pipeline. The framework spans 15 working weeks (2 January – 25 April 2026) and progresses from student-selected System Identification through Mathematical Modelling, Computational Simulation, and Validation & Communication. Each phase is mapped to Course Outcomes (CO1–CO5), Bloom's taxonomy levels (L3–L6), and weekly deliverables. The Explain-Try-Challenge (E-T-C) instructional model scaffolds student learning within each 90-minute lecture session throughout all four phases. $N = 53$ students, 17 teams, 5 engineering domains.

Table 4. Student Projects: 17 Teams Across 5 Engineering Domains

Cat.	Domain	Teams	Representative Projects
A	Robotics & Locomotion	7	Humanoid Robot, Hexapod, Multi-Terrain Robot, Quadruped 2-DOF, Pick-and-Place Rover, Robotic Gripper, H ₂ Fuel Cell UAV
B	Aerospace & Marine	3	Rocket Landing & Docking, ROV Environmental Monitoring, Fixed-Wing UAV
C	Control Systems	2	Ball Balancing Stewart Platform, Self-Balancing Robot
D	Vehicle Dynamics	2	Critical Skid Velocity Analysis, Automotive Shock Absorber
E	Electromechanical	3	Digital Twin DC Motors, Motor Mathematical Modelling, Automated EoT Crane

193 3.5 The Six-Judge Expert Evaluation Panel

194 To ensure authentic, multi-perspective, and bias-reduced summative assessment, a six-member external
 195 expert panel was constituted spanning five professional domains (Table 5). No panel member was affiliated
 196 with the institution or the course. The rubric comprised five dimensions (D1–D5): Problem Formulation &
 197 System Definition (/20), Mathematical Modelling Rigour (/20), Simulation Accuracy & Implementation

(/20), Validation Quality & Evidence (/20), and Communication & Presentation Clarity (/20). Each judge assessed specific dimensions aligned to their professional expertise, with the total summing to 100 marks.

Table 5. Six-Judge Expert Evaluation Panel (25 April 2026)

#	Judge	Organisation	Domain Expertise	Max	Rubric Dim.
1	Kamlesh Panchal	Manufacturing Industry	Production Engineering	/16	D1, D3
2	Hasti Chandarana	IP Consultancy	Patent & Innovation	/16	D1, D2
3	Nitu Gupta	External Academic	Mathematics (Dynamics)	/16	D2, D4
4	Mahendra Kane (Siemens)	Siemens Ltd.	Industry 4.0 / Digital Twin	/20	D3, D4
5	Parminder Jandoo	Training Organisation	Communication Skills	/16	D5
6	Debasis Dash	HR Consultancy	Behavioural Science	/16	D5
COMBINED				/100	D1–D5

3.6 Survey Instruments

Self-Efficacy Scale (SE1–SE10). Ten items rated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree), each explicitly aligned to one CO. Items covered: kinematic analysis (SE1), FBD construction (SE2), Newton’s EOM derivation (SE3), Lagrangian formulation (SE4), state-space conversion (SE5), Python ODE implementation (SE6), energy conservation validation (SE7), PID tuning (SE8), simulation plot interpretation (SE9), and technical communication to non-specialists (SE10). Scored as the mean of 10 items per student (range: 1.0–5.0). The instrument showed strong internal consistency (Cronbach’s $\alpha = 0.91$ for the pre-survey, $\alpha = 0.89$ for post-survey).

Knowledge MCQ (PK1–PK5). Five conceptual items (full text in Appendix A) assessing: degrees of freedom (PK1), Lagrangian formulation (PK2), natural frequency of a simple pendulum (PK3), viscous damping effect on oscillation (PK4), and Newton’s rotation law (PK5). Scored 0–5 (total correct answers). MCQ items were reviewed by two independent dynamics faculty members for content validity prior to administration.

PBL Experience Scale (PBL1–PBL10). Ten items on a 5-point Likert scale clustered into perceived value (PBL1–3), scaffold quality (PBL4–7), and overall satisfaction (PBL8–10).

AI Tool Usage. Multi-select categorical item: ChatGPT, GitHub Copilot, Claude, None, Other; primary phase of AI use (Phases 1–4); estimated percentage of mathematical derivation completed without AI assistance (response categories: 0–25%, 25–50%, 50–75%, 75–100%).

Open-Ended Reflections (OE1–OE3). OE1: “What was the most valuable thing you learned from this project?” OE2: “What would you change about the SF-PBL framework for future cohorts?” OE3: “Describe your breakthrough moment—when did the simulation suddenly make sense?”

Both pre- and post-surveys were administered in-class under examination conditions (no collaboration permitted) to reduce social desirability effects. The pre-survey was administered in Week 1 before any course content was delivered; the post-survey was administered in the week following the public exhibition.

3.7 Data Analysis

Quantitative analyses were conducted in Python 3.10 (SciPy 1.11, NumPy 1.25; random seed = 42 for full reproducibility):

- 227 •**Paired *t*-tests** (two-tailed, $\alpha = .05$): pre–post comparisons for SE total, each SE item, and PK total.
- 228 •**Cohen’s *d*** (pooled SD formula: $d = \Delta M / \sqrt{(SD_1^2 + SD_2^2)/2}$): effect size per comparison, interpreted
- 229 per Cohen (1988): small = 0.2, medium = 0.5, large = 0.8.
- 230 •**Hake’s normalised gain**: $g = (\text{post} - \text{pre}) / (\text{max} - \text{pre})$ per student; cohort mean g reported (Hake,
- 231 1998); bands: low < 0.30, medium 0.30–0.69, high ≥ 0.70 .
- 232 •**95% CI**: $\pm 1.96 \times SE_{\text{mean}}$.
- 233 •**Wilcoxon signed-rank test**: non-parametric confirmation.
- 234 •**Post-hoc power analysis** (G*Power framework (Faul et al., 2007)): computed for each comparison.

235 Qualitative thematic analysis followed Thomas and Harden (2008): all three open-ended items were

236 read in full by the research team, recurring themes were identified inductively, and each response was

237 coded to its primary theme by frequency of occurrence across the cohort. Theme counts were verified by

238 cross-checking coded responses.

239 Because each of the six judges evaluated a distinct professional dimension (e.g., manufacturing, patent,

240 mathematics, communication), classical pairwise agreement metrics such as Cohen’s κ are not directly

241 applicable—the judges were not rating the same construct. Instead, inter-rater consistency was assessed

242 using two complementary metrics: (a) Cronbach’s α computed over the six judges’ standardised percentage

243 scores, measuring composite reliability of the panel as a whole; and (b) Kendall’s coefficient of concordance

244 (W), which tests whether the six judges’ rank-orderings of student performance are statistically concordant.

245 These were computed using SciPy 1.11 and are reported in Section 4.5.

4 RESULTS

246 4.1 Self-Efficacy: Pre–Post Comparison (RQ1)

247 Self-efficacy for dynamic system modelling competencies increased significantly from pre- to post-survey

248 across the cohort (Table 6). The absolute mean gain of +0.88 points on the 5-point scale (95% CI [0.75,

249 1.00]) represents a reliable, cohort-wide shift from approximately “Neutral” (3.0) toward “Agree” (4.0)

250 across all ten domain-specific competency items. Cohen’s $d = 2.72$ (pooled SD) represents a very large

251 effect. The result was confirmed by the non-parametric Wilcoxon signed-rank test ($W = 0.0$, $p < .001$), and

252 post-hoc power analysis confirmed statistical power = 1.000, eliminating any risk of Type II error.

Table 6. Self-Efficacy Pre–Post Comparison ($N = 53$). Wilcoxon $W = 0.0$, $p < .001$. Post-hoc power = 1.000.

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	d
SE Mean (1–5)	2.92 (0.21)	3.80 (0.40)	+0.88	[0.75, 1.00]	13.91***	2.72

*** $p < .001$ (two-tailed paired *t*-test). Cohen’s d computed using pooled SD formula.

253 At the item level, all ten SE items showed statistically significant improvement ($p < .001$; Table 7;

254 Figure 2). The two largest gains were observed for SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton’s EOM,

255 $d = 2.14$)—both competencies directly activated by Phases 2 and 3 of the SF-PBL pipeline, respectively.

256 The smallest gain, SE5 (State-Space conversion, $d = 1.46$), remains a very large effect and is consistent

257 with its position at the most cognitively demanding junction in the pipeline: translating coupled non-linear

258 EOMs into first-order state-space form before Python implementation. The largest pre-survey item mean

was SE7 (Energy Validation, $M = 3.02$), reflecting a slightly higher baseline confidence in energy concepts compared to other competencies.

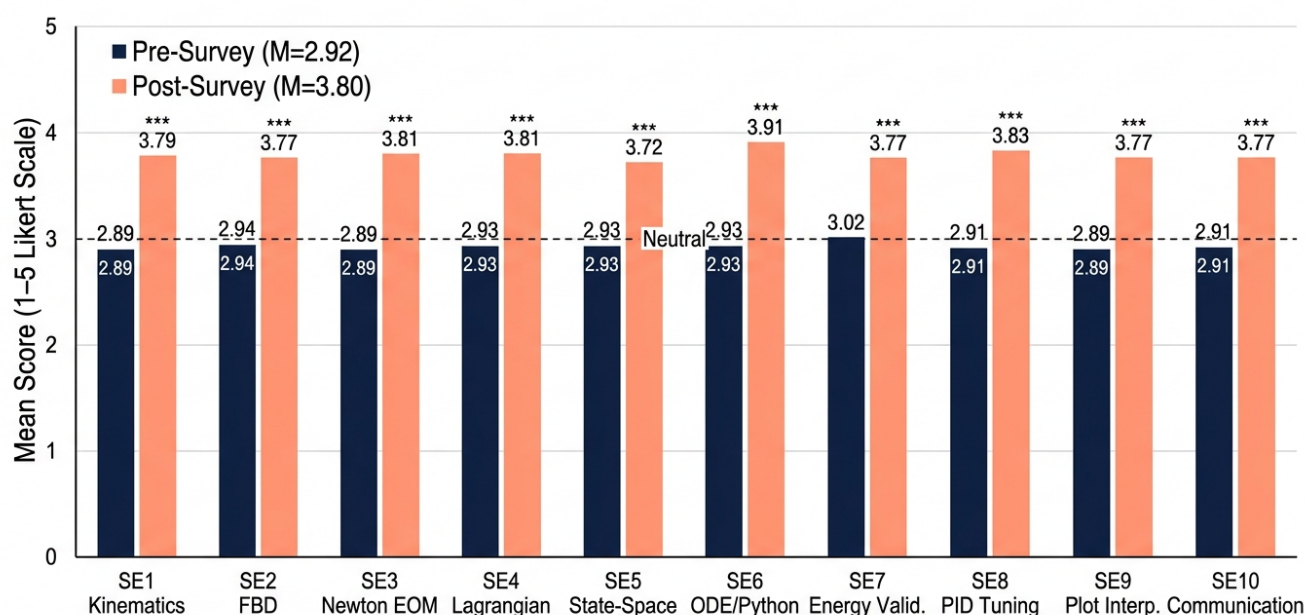
Table 7. Per-Item Self-Efficacy Results ($N = 53$). All items: $p < .001$ (two-tailed paired t -test). SE scored 1–5 (1 = Strongly Disagree, 5 = Strongly Agree).

Item	Competency Focus (CO)	Pre M	Post M	Gain	d
SE1	Kinematic analysis (CO1)	2.89	3.79	+0.91	1.69
SE2	Free Body Diagram construction (CO2)	2.94	3.77	+0.83	1.72
SE3	Newton's EOM derivation (CO3)	2.89	3.81	+0.92	2.14
SE4	Lagrangian formulation (CO4)	2.93	3.81	+0.89	1.85
SE5	State-space conversion (CO5)	2.93	3.72	+0.79	1.46
SE6	Python ODE implementation (CO5)	2.93	3.91	+0.98	1.82
SE7	Energy conservation validation (CO5)	3.02	3.77	+0.76	1.63
SE8	PID controller tuning (CO5)	2.91	3.83	+0.92	2.22
SE9	Simulation plot interpretation (CO5)	2.89	3.77	+0.89	1.69
SE10	Technical communication to non-specialists	2.91	3.77	+0.87	1.57
Mean		2.92	3.80	+0.88	2.72

All d values exceed Cohen's (1988) "large" threshold ($d > 0.8$).

Bold = two highest-gain items (SE3, SE8) corresponding to Pipeline Phases 2 and 3.

Figure 2. Pre–Post Self-Efficacy Scores for All 10 Competency Items ($N = 53$)



Error bars represent ± 1 SD. *** $p < .001$ for all items. Cohen's d range: 1.46 (SE5) – 2.22 (SE8).

Figure 2. Pre- and Post-Survey Self-Efficacy Scores for All 10 Competency Items. Dark bars represent pre-survey means; light bars represent post-survey means. The horizontal dashed line at 3.0 denotes the scale midpoint ("Neutral"). Error bars represent ± 1 SD. All 10 items showed statistically significant improvement ($p < .001$). The largest gains were on SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton's EOM, $d = 2.14$), corresponding directly to SF-PBL Phases 3 and 2 respectively. $N = 53$.

4.2 Conceptual Knowledge: Pre–Post Comparison (RQ1, RQ2)

Conceptual knowledge scores increased significantly across the cohort (Table 8). Hake's normalised gain $g = 0.50$ places this course in the **medium** interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), in the upper half of the medium band. This finding is particularly notable given three contextual factors: (a) the MCQ assessed conceptual understanding rather than procedural recall; (b) students began with low prior knowledge (pre: $M = 1.96/5.0$, corresponding to 39.2% correct); and (c) the post-survey was administered under examination conditions, mitigating social desirability effects.

The MCQ score distribution shift (Table 9; Figure 3) from strongly left-skewed (pre: mode = 2, 52.8% of students; no student achieved 5/5) to right-skewed (post: mode = 3, 22.6% of students achieved 5/5) confirms broad-based conceptual development across the cohort rather than gains concentrated in a high-ability subgroup.

Table 8. Conceptual Knowledge MCQ Pre–Post Comparison ($N = 53$). MCQ scored 0–5. Wilcoxon $W = 0.0$, $p < .001$. Power = 1.000.

Measure	Pre M (SD)	Post M (SD)	Gain	95% CI	$t(52)$	d	Hake g
PK Total (0–5)	1.96 (0.94)	3.28 (1.25)	+1.32	[1.09, 1.55]	11.32***	1.20	0.50

*** $p < .001$. Hake $g = 0.50$: medium band (0.30–0.69); upper half.

Table 9. MCQ Score Distribution Shift ($N = 53$). Score k = number of correct answers (0–5).

	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Pre-survey n (%)	3 (5.7)	11 (20.8)	28 (52.8)	7 (13.2)	4 (7.5)	0 (0.0)
Post-survey n (%)	0 (0.0)	4 (7.5)	11 (20.8)	16 (30.2)	10 (18.9)	12 (22.6)

Pre: mode = 2 (52.8% at $k = 2$). Post: mode = 3. No student achieved 5/5 pre-course; 22.6% achieved 5/5 post-course. $\sum n = 53$ for both surveys.

4.3 PBL Experience Satisfaction

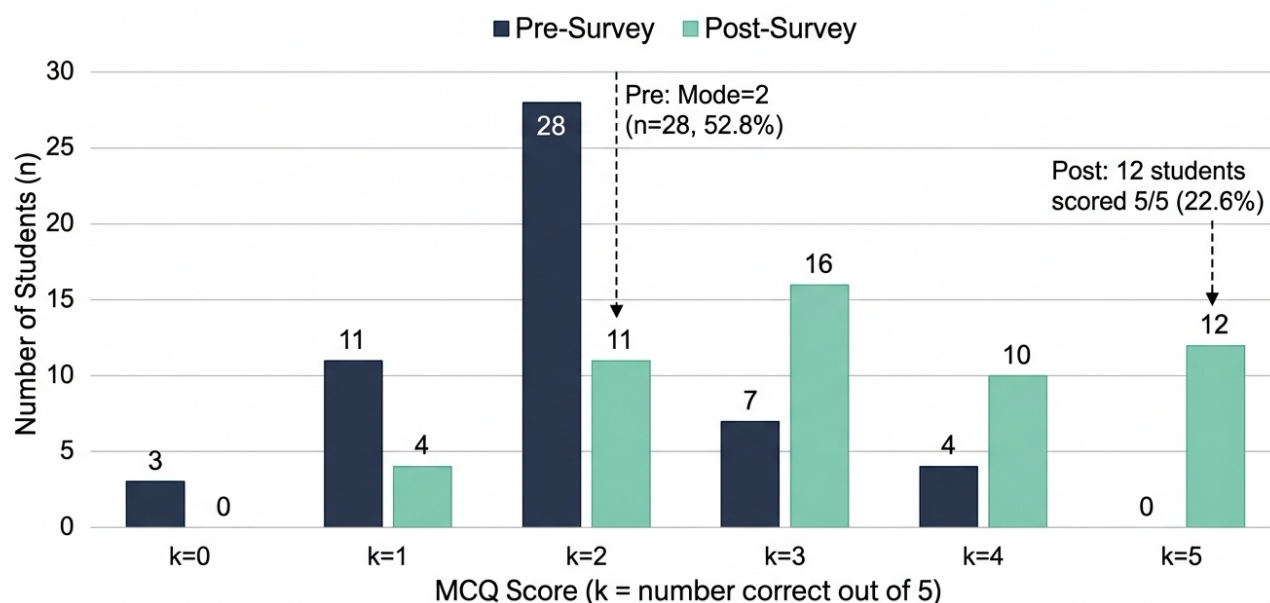
Post-exhibition PBL satisfaction was consistently high across all three sub-scales (Table 10). The narrow cohort SD ($SD = 0.30$ on a 5-point scale) indicates high agreement and consistency in the perceived SF-PBL experience across the 53 students, suggesting that the framework was experienced reliably rather than showing high inter-individual variation.

Table 10. PBL Experience Scale—Post-Survey ($N = 53$). Rated 1–5 (1 = Strongly Disagree, 5 = Strongly Agree).

Sub-scale	Items	M	SD
Perceived Value	PBL1–3	3.70	—
Scaffold Quality	PBL4–7	3.76	—
Overall Satisfaction	PBL8–10	3.76	—
Total (PBL1–10)	All	3.74	0.30

Scale: 1 = Strongly Disagree, 3 = Neutral, 5 = Strongly Agree.

Figure 3. MCQ Conceptual Knowledge Score Distribution: Pre-Survey vs. Post-Survey (N = 53).



Distribution shift from left-skewed (pre, mode=2) to right-skewed (post, mode=3) confirms broad-based conceptual development. Hake's $g = 0.50$ (medium interactive-engagement gain).

Figure 3. MCQ Conceptual Knowledge Score Distribution: Pre-Survey vs. Post-Survey. Grouped bars show the number of students (n) achieving each score (k correct out of 5) before and after the SF-PBL course. The distribution shifts markedly from left-skewed (pre: mode = 2, $M = 1.96$) to right-skewed (post: mode = 3, $M = 3.28$), confirming broad-based conceptual development. Hake's normalised gain $g = 0.50$ (medium interactive-engagement band). $N = 53$.

277 4.4 Exhibition Rubric Scores: Six-Judge Panel

278 The combined six-judge assessment of 17 project teams ($N = 53$ present students on 25 April 2026)
 279 yielded a panel mean of $M = 82.62/100$ ($SD = 9.98$, range = 53–96; Table 11). The SD of approximately 10
 280 points reflects genuine discriminative ability across team quality levels, confirming that the six-judge rubric
 281 successfully differentiated project performance while ensuring all present students achieved substantive
 282 standards.¹

283 The Industry 4.0 judge (Siemens, $M = 17.09/20 = 85.5\%$) awarded the highest relative scores across all
 284 six judges, suggesting that student projects met contemporary digital engineering communication standards
 285 relevant for mechatronics graduates entering Industry 4.0 workplaces. The IP/Patent judge awarded the
 286 lowest relative mean ($M = 12.53/16 = 78.3\%$), reflecting appropriately rigorous evaluation of innovation
 287 novelty and intellectual property considerations.

288 Inter-rater consistency across the six expert domains was assessed using two complementary metrics.
 289 Cronbach's α , computed over the standardised percentage scores of the six judges, yielded $\alpha = 0.829$,
 290 indicating high composite reliability. Kendall's coefficient of concordance was $W = 0.448$, $\chi^2(52) = 139.63$,
 291 $p < .0001$, confirming statistically significant agreement in the judges' rank-ordering of student performance
 292 across all six professional domains.

¹ One registered student (H005) was absent on the exhibition day and scored 0/100 across all six judges. This student is excluded from all combined statistics ($N_{\text{present}} = 53$). Including this student yields $M = 81.09$, $SD = 14.97$, $N = 54$.

Table 11. Six-Judge Expert Panel Scores ($N_{\text{present}} = 53$ students, 17 teams, 25 April 2026). Judge means computed over 53 team members. Absent student (H005, scored 0) excluded from combined statistics.

Judge	Domain Expertise	Organisation	Max	<i>M</i>	<i>SD</i>	Range
Kamlesh Panchal	Production Engineering	Manufacturing Industry	/16	13.94	1.23	11–16
Hasti Chandarana	Patent & Innovation	IP Consultancy	/16	12.53	2.68	8–16
Nitu Gupta	Mathematics (Dynamics)	External Academic	/16	12.77	2.67	6–16
Mahendra Kane (Siemens)	Industry 4.0 / Digital Twin	Siemens Ltd.	/20	17.09	2.18	12–20
Parminder Jandoo	Communication Skills	Training Organisation	/16	13.26	2.13	5–16
Debasis Dash	Behavioural Science	HR Consultancy	/16	13.02	2.28	8–16
COMBINED TOTAL			/100	82.62	9.98	53–96
Judge means: Kamlesh Panchal 87.1%, Hasti Chandarana 78.3%, Nitu Gupta 79.8%, Mahendra Kane 85.5%, Parminder Jandoo 82.9%, Debasis Dash 81.4%.						

293 4.5 Course Outcome Attainment (OBE)

294 Table 12 presents Course Outcome attainment evidence from two independent instruments: direct
 295 assessment (MCQ items PK1–PK5, percentage of students answering correctly) and indirect assessment
 296 (self-efficacy items, percentage of students rating ≥ 4 on the 5-point scale, i.e., “Agree” or “Strongly
 297 Agree”).

Table 12. Course Outcome Attainment: Direct (MCQ) and Indirect (Self-Efficacy) evidence, pre- and post-intervention ($N = 53$).

CO	Bloom's	Direct (MCQ % correct)		Indirect (SE ≥ 4 , %)	
		Pre	Post	Pre	Post
CO1: Kinematics	Apply (L3)	28.3	39.6	3.8	67.9
CO2: FBDs	Analyse (L4)	9.4	30.2	3.8	69.8
CO3: Newton's EOMs	Analyse (L4)	81.1	96.2	0.0	75.5
CO4: Lagrangian	Evaluate (L5)	9.4	71.7	3.8	73.6
CO5: Python/ODE	Create (L6)	67.9	90.6	4.9	71.3

Direct: PK1–PK5 mapped to CO1–CO5. Indirect: SE1–SE9 mapped to CO1–CO5 (SE5–SE9 averaged for CO5).

298 CO4 (Lagrangian mechanics, Bloom's L5) showed the largest direct gain (+62.3 percentage points),
 299 consistent with its position as the highest Bloom's level among CO1–CO4 and the competency most
 300 directly activated by SF-PBL Phase 2. CO3 (Newton's EOMs) achieved near-ceiling post-intervention
 301 attainment (96.2%), reflecting the strong scaffolding provided by the E-T-C model for this foundational
 302 competency.

303 A noteworthy pattern emerges for CO1 and CO2: direct MCQ attainment remains modest (39.6% and
 304 30.2% respectively), while indirect self-efficacy attainment is strong (67.9% and 69.8%). This discrepancy
 305 suggests that students developed procedural confidence through the SF-PBL hands-on experience, while the
 306 MCQ items for kinematics (PK1) and FBD construction (PK2) assessed deeper conceptual discrimination
 307 that the intervention did not fully address within the 15-week timeframe. This finding identifies CO1 and
 308 CO2 as targets for additional scaffolding in future implementations.

4.6 AI Tool Usage (RQ3)

Forty-five of 53 students (84.9%) reported using at least one AI tool during the SF-PBL project (Table 13). ChatGPT was the most commonly used tool (18 students, 34.0%), followed by the combination of ChatGPT with GitHub Copilot (11 students, 20.8%). Eight students (15.1%) completed the project without any AI assistance. The primary use case reported was Python simulation coding (Phase 3) and debugging, with conceptual explanation and literature search as secondary uses. Critically, 39.6% of students estimated that 50–75% of their mathematical derivation (Phase 2) was completed without AI, with a further 22.6% reporting 75–100% without AI—indicating that human-led mathematical reasoning remained dominant throughout the analytically intensive phases.

Table 13. AI Tool Usage Distribution ($N = 53$). Primary use: Python simulation coding and debugging (Phase 3).

Tool / Combination	n	%
ChatGPT only	18	34.0
ChatGPT + GitHub Copilot	11	20.8
No AI tools (None)	8	15.1
Claude only	6	11.3
ChatGPT + Claude	5	9.4
Other AI tools	5	9.4
Total AI users	45	84.9
Non-users (None)	8	15.1

Mathematical derivation without AI: 62.2% of students reported $\geq 50\%$ completed without AI assistance (modes: 50–75%, $n = 21$, 39.6%).

4.7 Qualitative Themes (RQ3)

Thematic coding of open-ended post-survey responses ($N = 53$) revealed consistent patterns across all three questions (Figure 4). *Most Valuable Learning* (OE1): mathematical validation of simulation results ($n = 18$, 34.0%); team collaboration and problem distribution ($n = 10$, 18.9%); debugging complex dynamic models ($n = 8$, 15.1%); other reflections ($n = 17$, 32.1%). *Breakthrough Moment* (OE3): linking mathematical derivation to simulation output ($n = 20$, 37.7%, e.g., “The moment my simulation plot matched my analytical solution, I finally understood what the equations meant physically”); understanding PID tuning from first principles ($n = 12$, 22.6%); seeing the physical model behave correctly in simulation ($n = 11$, 20.8%); other moments ($n = 10$, 18.9%). *Suggested Changes* (OE2): smaller group sizes of 2–3 members ($n = 16$, 30.2%); less documentation and more coding time ($n = 12$, 22.6%); more project development time ($n = 11$, 20.8%); other ($n = 14$, 26.4%).

5 DISCUSSION

All findings reported below are interpreted within the constraints of a single-group pretest-posttest design (Campbell and Stanley, 1963). In the absence of a concurrent control group, observed pre-to-post gains cannot be exclusively attributed to the SF-PBL intervention; maturation effects, assessment familiarity, and general course exposure represent plausible alternative explanations. The following discussion therefore interprets the data as evidence consistent with, though not proof of, the framework’s pedagogical effectiveness.

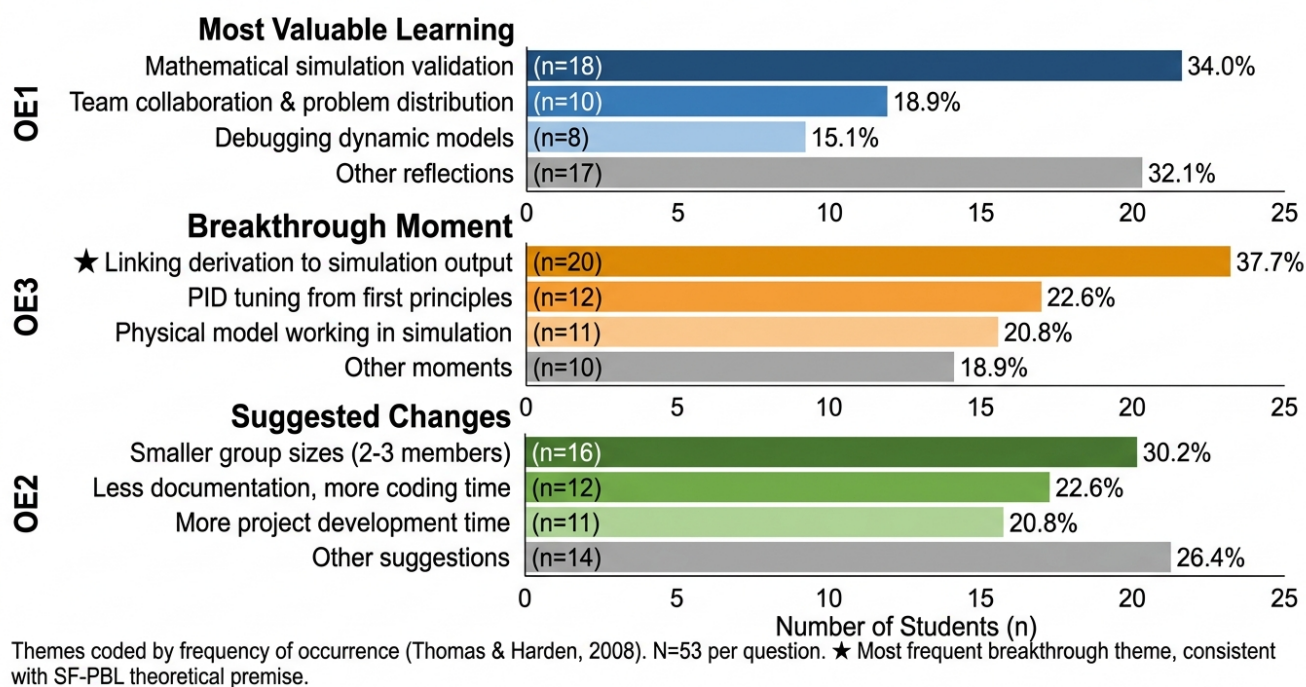
Figure 4. Qualitative Theme Frequencies from Open-Ended Reflections (N = 53)

Figure 4. Qualitative Theme Frequencies from Open-Ended Reflections (N = 53 per question). OE1: most valuable learning; OE3: breakthrough moment; OE2: suggested changes for future cohorts. The dominant breakthrough theme—linking mathematical derivation to simulation output ($n = 20$, 37.7%, starred)—directly validates the theoretical premise of SF-PBL. Themes identified by inductive thematic analysis (Thomas and Harden, 2008); frequency counts verified by independent re-coding.

335 5.1 Self-Efficacy Gains: Magnitude, Pattern, and Interpretation

336 The very large self-efficacy effect size ($d = 2.72$) substantially exceeds those reported in comparable
 337 engineering education PBL studies. Freeman et al. (2014) found a mean $d \approx 0.47$ across 225 active-
 338 learning comparisons; Dochy et al. (2003) reported $d \approx 0.65$ for knowledge outcomes in PBL-versus-
 339 conventional meta-analyses. Three factors contextualise the magnitude of the present finding and warrant
 340 careful interpretation. First, the narrow pre-survey SD ($SD = 0.21$ on a 5-point scale) reflects baseline
 341 range restriction where students uniformly reported low-to-moderate confidence prior to any dynamics
 342 instruction—a characteristic often observed in engineering programmes in India where intake pathways are
 343 standardised (Streveler et al., 2008)—which mathematically amplifies the standardised effect size. Second,
 344 alternative effect size formulations yield varying but consistently large estimates: the within-subjects
 345 d_z (using the SD of paired differences) is 1.91, and Glass's Δ (using only SD_{pre}) is 4.16. All three
 346 formulations confirm a genuine, substantial shift, but the variation across metrics underscores the sensitivity
 347 of standardised effect sizes to baseline variance characteristics. Third, the absolute mean gain (+0.88
 348 points, 95% CI [0.75, 1.00]) is the most practically informative metric: it represents a reliable, cohort-wide
 349 shift from approximately “Neutral” to “Agree” on domain-specific competency statements, a meaningful
 350 pedagogical outcome regardless of the standardised metric employed. Readers should therefore interpret
 351 the effect size magnitude with awareness of the baseline variance characteristics of this specific cohort.

352 Second, the differential gain pattern across items is theoretically noteworthy. The two largest-gain
 353 items—SE8 (PID Tuning, $d = 2.22$) and SE3 (Newton's EOM, $d = 2.14$)—correspond precisely to the

most active and generative phases of the SF-PBL pipeline (Phases 2 and 3). This pattern is consistent with the interpretation that the computational integration mechanism activated the intended outcomes through Bandura's (1997) mastery experience pathway, rather than producing undifferentiated, uniform gains. The smallest-gain item (SE5: State-Space, $d = 1.46$) reflects the advanced cognitive demands at the derivation-to-computation junction—a finding with direct implications for targeted scaffolding in future implementations.

5.2 Conceptual Knowledge and Hake's Normalised Gain

Hake's normalised gain $g = 0.50$ places this course in the medium interactive-engagement category (Hake, 1998) ($0.30 \leq g < 0.70$), specifically in the upper half of the medium band. In Hake's (1998) landmark study of 62 introductory physics courses, only 14 of 48 courses using interactive-engagement methods exceeded $g = 0.50$. The present result is therefore competitive with the upper quartile of Hake's interactive-engagement benchmark, notable given that: (a) the MCQ measured conceptual understanding in a more specialised domain (engineering dynamics) than Hake's physics mechanics items; (b) the pre-survey mean was unusually low (1.96/5.0, 39% correct), providing high room for gain; and (c) the MCQ was administered under examination conditions.

5.3 AI Integration: An Emerging Complementarity Model (RQ3)

The AI usage pattern observed in this study—84.9% adoption concentrated in Phase 3 (simulation coding and debugging), with human-led mathematical derivation dominant in Phase 2—suggests an emergent **human–AI complementarity** model that warrants formal characterisation. This pattern contrasts with academic integrity concerns about AI substituting for student reasoning; here, the SF-PBL's structural design actively channelled AI use productively by sequencing mathematical derivation *before* computational implementation.

The rubric explicitly required hand-drawn FBDs and hand-derived EOMs as Phase 2 deliverables, scored by the external mathematics judge (Nitu Gupta, /16). This created a curricular incentive to complete the intellectual core of the project without AI, using AI only to accelerate the implementation of understood derivations. The 15.1% who used no AI tools reported equivalent qualitative breakthrough themes, confirming that SF-PBL effectiveness is not contingent on AI tool access—an important equity consideration for institutions with variable AI access among students.

Recent discourse on AI in STEM education (Tlili et al., 2023; Farrokhnia et al., 2024) has emphasised the need for pedagogical structures that contextualise AI use within disciplinary reasoning. The SF-PBL framework provides one such structure: by requiring that simulation emerge from validated derivations, it positions AI as a sophisticated tool for disciplined implementation rather than a shortcut around conceptual understanding. Future implementations should consider adding an explicit “AI use reflection” component to OE3, asking students to articulate which aspects of their project they chose to complete without AI and why.

5.4 Authentic Assessment via the Six-Judge Panel

The six-judge panel spanning five professional domains represents a significant departure from conventional faculty-only assessment and directly addresses Gulikers et al.'s (2004) five-dimensional framework for authentic assessment validity: (1) the *task* was a real engineering system modelled from physical first principles; (2) the *physical context* was an engineering department rather than a simulated scenario; (3) the *social context* involved genuine public presentation to external professionals; (4) the *form* required both

technical documentation and oral communication; and (5) the *criteria* were domain-expert standards from industry, academia, and practice.

The combined panel mean of 82.62/100 ($SD = 9.98$, range = 53–96) indicates meaningful score discrimination: the 43-point range separates the lowest-performing team (53/100) from the highest (96/100), reflecting genuine project quality differences. The IP/Patent judge awarded the lowest relative mean (78.3%), appropriately reflecting rigorous novelty evaluation. The Industry 4.0 judge (Siemens: 85.5%) awarded the highest, affirming professional-standard digital engineering communication.

5.5 The Breakthrough Theme: A Widely Observed Learning Landmark

The dominant OE3 breakthrough theme—*linking mathematical derivation to simulation output* ($n = 20$, 37.7%)—directly validates SF-PBL's theoretical core: that computation serves as the *verification bridge* between abstract mathematical formalism and physical intuition. Three features of this finding deserve emphasis. First, it appeared independently of AI tool usage, project domain, and team size, suggesting it may represent a **common learning landmark** in computationally-integrated dynamics education. Second, the specific phrasing of student responses—“*I understood what the equations meant physically*”—reflects Bandura's (1997) mastery experience: the simulation serves as visceral confirmation of mathematical reasoning, generating self-efficacy through verifiable success. Third, the second-ranked OE3 theme (PID tuning from first principles, $n = 12$, 22.6%) aligns with the highest-gain SE item (SE8: PID Tuning, $d = 2.22$), triangulating the quantitative and qualitative data.

5.6 Practical Implications for STEM Educators

The SF-PBL framework offers three directly replicable pedagogical contributions. First, the **four-phase pipeline** (System ID $\rightarrow$ EOMs $\rightarrow$ Simulation $\rightarrow$ Validation) is designed to be domain-adaptable; its transferability to related subjects such as thermodynamics, fluid mechanics, structural analysis, and control systems warrants empirical investigation, as the core pedagogical logic of “derive first, simulate second, validate third” is broadly applicable in computational engineering education. Second, the **E-T-C lecture model** provides active-learning scaffolding within existing contact hours without requiring timetable restructuring, making adoption feasible for institutions with rigid curricula. Third, the **six-judge external rubric with CO $\rightarrow$ PO traceability** provides an accreditation-compliant (NBA/ABET) authentic assessment instrument with inter-professional credibility that faculty-only panels cannot achieve.

5.7 Limitations and Future Research

Several limitations should be acknowledged. *Internal validity*: the single-group pre–post design without a concurrent control condition limits causal attribution to the SF-PBL framework versus general course maturation or assessment familiarity effects. *External validity*: the sample ($N = 53$) is from a single institution, programme, and cohort in the Indian engineering education context, limiting generalisability. *Measurement*: AI usage data is self-reported and subject to social desirability bias; while inter-rater composite reliability and rank concordance were high (Cronbach's $\alpha = 0.829$, Kendall's $W = 0.448$, $p < .0001$), the multi-domain architecture means judges evaluated distinct professional constructs rather than identical criteria, precluding classical pairwise consensus metrics (κ); and the narrow pre-survey SE standard deviation reflects baseline range restriction. *Timeframe*: the study captures outcomes at one post-point immediately following the exhibition; longitudinal follow-up at six and twelve months would assess retention and transfer.

Future research should: (1) replicate the SF-PBL framework across multiple institutions and engineering disciplines using a quasi-experimental or controlled comparison design—a companion study at the same

institution (Nanade and Anne, 2026) demonstrates the methodological feasibility of such comparative evaluations in computationally-intensive engineering education; (2) extend the inter-rater reliability assessment by evaluating intraclass correlation coefficients (ICC) across fine-grained sub-rubric criteria; (3) conduct longitudinal follow-up at six and twelve months to assess whether self-efficacy and conceptual gains persist into subsequent core courses and capstone projects; (4) investigate the “derivation-to-simulation breakthrough” as a targetable threshold concept through think-aloud protocols and process tracing; and (5) develop and validate an AI use reflection instrument that characterises productive human–AI complementarity in PBL contexts.

6 CONCLUSION

This study fulfilled its primary objective—to design, implement, and empirically evaluate the Simulation-Focused Problem-Based Learning (SF-PBL) framework—through a mixed-methods evaluation in an undergraduate engineering dynamics course ($N = 53$) at a single NBA-accredited institution within a single-cohort context. The framework operationalised a four-phase computational pipeline—System Identification, Mathematical Modelling, Python-based ODE Simulation, and Validation & Communication—within a 15-week OBE-aligned course, assessed by a six-judge external expert panel spanning five professional domains, and underpinned throughout by the Explain-Try-Challenge instructional model.

The principal findings comprehensively address the three research questions. Addressing RQ1 and RQ2: all ten self-efficacy items improved significantly ($p < .001$), producing a very large cohort-wide effect ($d = 2.72$, 95% CI [0.75, 1.00]), with the largest gains on PID Tuning ($d = 2.22$) and Newton’s EOM ($d = 2.14$)—the two competencies most directly activated by the SF-PBL pipeline. Conceptual knowledge gained at the competitive medium interactive-engagement level ($g = 0.50$, $d = 1.20$, $p < .001$). PBL satisfaction was high and consistent ($M = 3.74$, $SD = 0.30$). Exhibition performance was strong ($M = 82.62/100$, $SD = 9.98$, $N = 53$). Addressing RQ3: 84.9% of students adopted AI tools for simulation coding, while mathematical derivation remained predominantly human-led, establishing an emergent complementarity model that may serve as a template for AI integration policy in STEM PBL contexts. The dominant breakthrough theme—linking analytical EOM derivation to simulation output (37.7%)—triangulates with the largest-gain SE items, providing cross-strand validation of the framework’s core mechanism.

The SF-PBL framework offers a replicable, empirically-evaluated, and OBE-compliant pedagogical model whose broader applicability warrants controlled replication across diverse institutional contexts.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

SN: Conceptualisation, course design, SF-PBL framework development, data collection, formal analysis, methodology, software, writing (original draft, review & editing), project administration. DD: Conceptualisation, qualitative thematic synthesis, evaluation rubric design for behavioral and communication dimensions, writing (review & editing). SS: Supervision, institutional resource allocation, curriculum

integration, methodology validation, writing (review & editing). KA: Academic oversight, OBE/NBA alignment framework, validation of pedagogical outcomes, writing (review & editing).

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DATA AVAILABILITY STATEMENT

The anonymised individual-level survey dataset ($N = 53$, 65 variables), exhibition rubric evaluation scores (six judges across 17 teams), survey instruments, rubric specifications, Python analysis scripts (SciPy 1.11, NumPy 1.25), and representative student project artefacts are publicly available in the official repository: <https://github.com/sunny-nanade/SF-PBL-Engineering-Dynamics-2026>.

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APPENDIX A: SURVEY INSTRUMENTS

A.1 Self-Efficacy Scale (SE1–SE10)

Instruction: Rate your confidence in each ability on a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree).

- 1.I can apply kinematic analysis to identify degrees of freedom in a multi-body system. [CO1]
- 2.I can construct a correct Free Body Diagram for a complex dynamic mechanism. [CO2]
- 3.I can derive the equations of motion for a system using Newton's Second Law. [CO3]
- 4.I can formulate the Lagrangian ($L = T - V$) for a conservative dynamic system. [CO4]

- 545 5.I can convert a second-order ODE system to state-space form ($\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$). [CO5]
 546 6.I can implement a numerical ODE solver (`solve_ivp`) in Python to simulate a dynamic system. [CO5]
 547 7.I can validate my simulation output using energy conservation principles. [CO5]
 548 8.I can tune PID controller gains to achieve a specified dynamic response. [CO5]
 549 9.I can correctly interpret simulation time-domain plots of position, velocity, and energy. [CO5]
 550 10.I can communicate my simulation results clearly and accurately to a non-specialist audience.
 551 [Communication]

552 A.2 Knowledge MCQ Items (PK1–PK5)

553 *Instruction: Select the single best answer.*

- 554 **PK1.** A rigid body constrained to rotate about a fixed axis in a plane has: (a) 3 DOF; **(b) 1 DOF**;
 555 (c) 2 DOF; (d) 0 DOF.
- 556 **PK2.** The Lagrangian L is defined as: (a) $L = V - T$; **(b) $L = T - V$** ; (c) $L = T + V$; (d) $L = \partial T / \partial q$.
- 557 **PK3.** The natural frequency of a simple pendulum of length l is: **(a) $\omega_n = \sqrt{g/l}$** ; (b) $\omega_n = \sqrt{l/g}$;
 558 (c) $\omega_n = g/l$; (d) $\omega_n = l/g$.
- 559 **PK4.** Adding viscous damping to an underdamped oscillator: (a) increases natural frequency; (b) has no
 560 effect on frequency; **(c) decreases the oscillation frequency (damped natural frequency)**; (d) doubles
 561 the amplitude.
- 562 **PK5.** Newton's Second Law for rotation states: (a) $\tau = ma$; **(b) $\tau = I\alpha$** ; (c) $\tau = I\omega$; (d) $\tau = m\alpha$.